

# TASHKENT APARTMENT PRICE PREDICTOR

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## PROJECT IN ONE LINE

Estimate a current August 2026 Tashkent apartment asking price from observable listing attributes.

Reference estimate — not an appraisal

4,214

MODELING ROWS

RF

SELECTED MODEL

0.681

TEST  $R^2$

\$27,195

TEST MAE

# WHO NEEDS IT — AND WHAT DOES ML DO?

Lock the user, input, output, and boundary in under 45 seconds.

## ● USER

Buyers, sellers, agents, and analysts who need a consistent August 2026 reference estimate.

## ● SUPERVISED REGRESSION

Inputs: district, size, rooms, apartment level, building levels, new-build/resale.

Target: asking price.

## ● OUTPUT + BOUNDARY

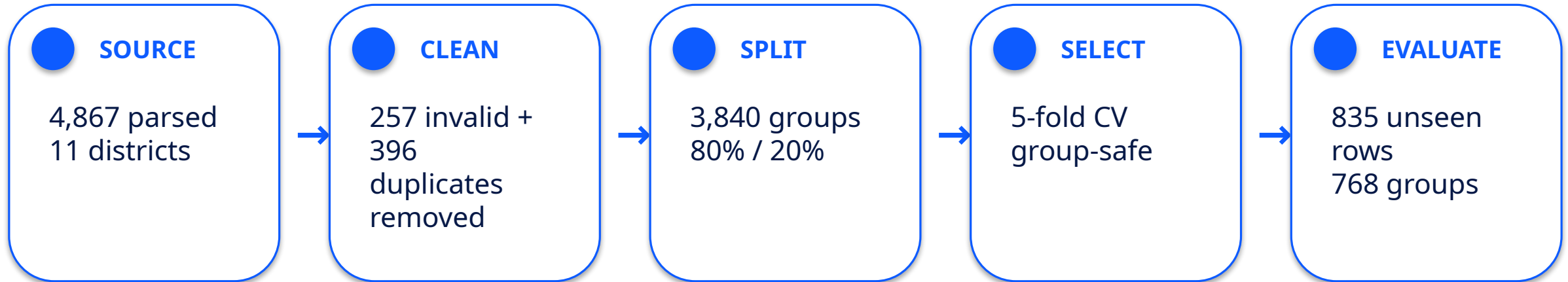
One estimated August 2026 asking price in USD, plus warnings.

Not a completed sale price, guarantee, or legal appraisal.

RAW APARTMENT → VALIDATED PIPELINE → USD ESTIMATE +  
WARNING

# FROM 4,867 LISTINGS TO A GROUP-SAFE TEST

Privacy-minimized public HATA snapshot from August 2026.



## Leakage controls

feature+target deduplication • identical fingerprints grouped • no price/m<sup>2</sup> leakage • preprocessing inside CV • test not used for selection

# FOUR APPROACHES — ONE EVIDENCE-BASED CHOICE

Final model selected by the lowest five-fold development CV MAE, not test performance.

| MODEL                | CV MAE          | CV RMSE         | CV R <sup>2</sup> |
|----------------------|-----------------|-----------------|-------------------|
| Median baseline      | \$55,604        | \$123,587       | -0.068            |
| Log Ridge            | \$38,301        | \$105,755       | 0.219             |
| <b>Random Forest</b> | <b>\$31,298</b> | <b>\$80,394</b> | <b>0.556</b>      |
| Gradient Boosting    | \$33,545        | \$94,081        | 0.396             |



## WHY RANDOM FOREST?

Best validation MAE

Captures non-linear size  
× location interactions

Trade-off: larger and less  
interpretable than Ridge

# UNSEEN-TEST RESULT — WITH THE FAILURE VISIBLE

The baseline comparison and error case matter more than a polished success claim.

**\$27,195**

MAE

**\$58,887**

RMSE

**0.681**

$R^2$

**24.58%**

MAPE

**−46.6%**

MAE vs baseline



## LARGEST ERROR

Shayhontohur • 160 m<sup>2</sup> • 3 rooms • new build  
Actual: \$1,000,000 Predicted: \$234,461  
Absolute error: \$765,539



## WHAT IT TEACHES

Luxury premiums and condition are not observed. Mirobod and Shayhontohur MAE is high; some district slice  $R^2$  values are negative.

# ONE REAL INPUT → ONE VISIBLE RESULT

Colab route: setup → load saved pipeline → predict → show validation error.

## RAW INPUT

|               |                   |
|---------------|-------------------|
| District      | Chilonzor         |
| Size          | 70 m <sup>2</sup> |
| Rooms         | 3                 |
| Level         | 3 / 5             |
| Building type | Resale            |



## ESTIMATED AUGUST 2026 ASKING PRICE

**\$97,098 USD**

## EDGE CASE

**Level 9 / max 5 → clear ValueError**

Open: README Colab badge • Run all • Do not tour source code unless asked

# HONEST LIMITS — CLEAR NEXT STEP

A useful educational model is stronger when its boundary is explicit.

## LIMITATIONS

Dated asking-price snapshot  
No condition / exact building  
/ legal status  
Unequal district reliability  
Source noise and market  
drift

## SAFE USE

August 2026 reference only  
Compare with recent listings  
Require human review  
No lending, tax, or legal  
appraisal

## NEXT IMPROVEMENT

Collect verified transactions  
with exact neighborhood,  
condition, building year,  
renovation, and legal status;  
then use a later time  
holdout.

**Thank you. I am ready for one evidence-based question.**

# OFFICIAL CRITERIA → EXACT REPOSITORY PROOF

Open docs/capstone\_evidence\_matrix.md for the complete claim → location → why-proof mapping.

## 1 Problem

README + Project Brief

## 2 Data

data/README + EDA +  
src/data.py

## 3 Models

model\_comparison.csv +  
src/model.py

## 4 Evaluation

test metrics + errors +  
district slices

## 5 Delivery

demo.ipynb + saved joblib  
+ predict.py

## 6 Reproduce

README +  
clean\_run\_check + CI

## 7 Responsible

README limitations +  
data card

## 8 Defense

deck + pitch + question  
bank

**YELLOW** only where external approval/rehearsal evidence is still required.



# THE SAME PIPELINE TRAINS AND PREDICTS

Preprocessing is serialized with the estimator, preventing notebook-only inference drift.

## ● RAW INPUT

district • rooms • size • level •  
max\_levels • new-build/resale

## ● VALIDATE

required fields • positive values •  
level  $\leq$  max\_levels

## ● FEATURE

floor\_ratio = level / max\_levels

## ● PREPROCESS

OneHotEncoder district • numeric  
passthrough / scaling

## ● MODEL

selected Random Forest inside  
sklearn Pipeline

## ● OUTPUT

USD estimate + unseen/range  
warnings

Source proof: [src/data.py](#) + [src/model.py](#) + [artifacts/apartment\\_price\\_pipeline.joblib](#)

# ANSWER WITHOUT GUESSING

Direct answer → exact evidence → limitation or next step.

## ● WHY RANDOM FOREST?

Lowest group-safe CV MAE (\$31,298).

Open  
reports/model\_comparison.csv.  
Trade-off: size and interpretability.

## ● IS IT CURRENT?

Yes — dated Aug 2026 asking-price snapshot.

Open data/README.md.  
Not a permanently live or completed-sale model.

## ● BIGGEST FAILURE?

\$1m Shayhontohur actual vs ~\$234k predicted.

Open largest\_errors.csv.  
Missing luxury/condition signal.

**Full evidence-anchored answers: docs/defense\_question\_bank.md**